

Project Report

Project 7: Understanding Metric Spaces Through Distance Functions

Author Name(s):

Anvi Patchigolla

Prajnaa M

Nitin Sailpathi

Harshil Pansala

Mohan Pritam K

Srikar Nandagiri

Aditya Gupta

Under the guidance of:

Dr. G. D. Veerappa Gowda

Contents

Abstract	3
1 Introduction	3
2 Definition of a Metric Space	3
3 Exploring Metrics on $\mathbb{R}^2$	3
3.1 The Euclidean Metric (d_2)	4
3.2 The Taxicab Metric (d_1)	5
3.3 The Discrete Metric (d_0)	6
4 Examples of Distance Calculations	6
4.1 Example: Euclidean Distance	7
4.2 Example: Taxicab Distance	7
4.3 Example: Discrete Distance	7
5 Unit Balls in Different Metrics	7
6 Python Code Used for Plotting	9
7 Analysis of How Different Metrics Affect Topological Properties	10
7.1 Euclidean Metric	10
7.2 Taxicab Metric	10
7.3 Discrete Metric	10
7.4 Summary	11

Abstract

This project explores the fundamental concept of metric spaces by analyzing three distinct distance functions defined on the Euclidean plane, $\mathbb{R}^2$. We investigate the **Euclidean**, **Taxicab (or Manhattan)**, and **Discrete** metrics. For each metric, we provide its formal definition and a detailed proof verifying that it satisfies the four axioms of a distance function. Finally, we provide concrete examples of distance calculations. This foundational work sets the stage for a deeper analysis of the geometric and topological properties induced by each metric.

1 Introduction

In mathematics, a “distance” is a more flexible concept than the everyday ruler measurement. A metric space is a set equipped with a distance function, or **metric**, that formalizes this notion. The study of metric spaces provides a crucial bridge between the fields of geometry, which deals with shape and size, and topology, which studies properties of spaces that are preserved under continuous deformation.

This project aims to explore this relationship by defining several key metrics on the familiar set $\mathbb{R}^2$. By providing rigorous proofs, we will confirm that each of these functions is a valid metric, thereby establishing a solid framework for any future analysis of their properties.

2 Definition of a Metric Space

A metric space is a pair (X, d) , where X is a non-empty set and d is a function called a metric.

Definition. A function $d : X \times X \rightarrow \mathbb{R}$ is a **metric** on X if for all points $x, y, z \in X$, it satisfies the following four axioms:

- (i) **Non-negativity:** $d(x, y) \geq 0$
- (ii) **Identity of Indiscernibles:** $d(x, y) = 0 \iff x = y$
- (iii) **Symmetry:** $d(x, y) = d(y, x)$
- (iv) **Triangle Inequality:** $d(x, z) \leq d(x, y) + d(y, z)$

These axioms ensure that the function d behaves like a “reasonable” measure of distance.

3 Exploring Metrics on $\mathbb{R}^2$

We will now define and rigorously prove that three well-known functions are indeed metrics on the set $\mathbb{R}^2$. Let $x = (x_1, x_2)$, $y = (y_1, y_2)$, and $z = (z_1, z_2)$ be three points in $\mathbb{R}^2$.

3.1 The Euclidean Metric (d_2)

Definition: The Euclidean metric $d_2 : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined as:

$$d_2(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

Detailed Proof: We must verify the four axioms.

- (i) **Non-negativity:** For any real numbers, $(x_1 - y_1)^2 \geq 0$ and $(x_2 - y_2)^2 \geq 0$. Their sum is also non-negative. The square root of a non-negative number is, by definition, non-negative. Thus, $d_2(x, y) \geq 0$.
- (ii) **Identity of Indiscernibles:** First, assume $d_2(x, y) = 0$.

$$\sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} = 0$$

Squaring both sides gives $(x_1 - y_1)^2 + (x_2 - y_2)^2 = 0$. Since both terms are non-negative, their sum can only be zero if each term is individually zero. Therefore, $(x_1 - y_1)^2 = 0 \implies x_1 - y_1 = 0 \implies x_1 = y_1$. And, $(x_2 - y_2)^2 = 0 \implies x_2 - y_2 = 0 \implies x_2 = y_2$. Since $x_1 = y_1$ and $x_2 = y_2$, the points x and y are identical: $x = y$.

Conversely, assume $x = y$. Then $x_1 = y_1$ and $x_2 = y_2$.

$$d_2(x, y) = \sqrt{(x_1 - x_1)^2 + (x_2 - x_2)^2} = \sqrt{0^2 + 0^2} = 0$$

Thus, $d_2(x, y) = 0 \iff x = y$.

- (iii) **Symmetry:** We use the property that $(a - b)^2 = (b - a)^2$ for real numbers.

$$\begin{aligned} d_2(x, y) &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} \\ &= \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2} \\ &= d_2(y, x) \end{aligned}$$

Thus, $d_2(x, y) = d_2(y, x)$.

- (iv) **Triangle Inequality:** We need to prove $d_2(x, z) \leq d_2(x, y) + d_2(y, z)$. This is equivalent to:

$$\sqrt{(x_1 - z_1)^2 + (x_2 - z_2)^2} \leq \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2} + \sqrt{(y_1 - z_1)^2 + (y_2 - z_2)^2}$$

Let $u = x - y = (x_1 - y_1, x_2 - y_2)$ and $v = y - z = (y_1 - z_1, y_2 - z_2)$. Then $x - z = u + v = (x_1 - z_1, x_2 - z_2)$. The inequality becomes $d_2(x, z) = \|u + v\|_2$ and $d_2(x, y) + d_2(y, z) = \|u\|_2 + \|v\|_2$. We must prove $\|u + v\|_2 \leq \|u\|_2 + \|v\|_2$. This is the standard Minkowski inequality for $p = 2$, which is a known property of the Euclidean norm (derived from the Cauchy-Schwarz inequality).

Since d_2 satisfies all four axioms, it is a valid metric on $\mathbb{R}^2$.

3.2 The Taxicab Metric (d_1)

Definition: The Taxicab metric $d_1 : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined as:

$$d_1(x, y) = |x_1 - y_1| + |x_2 - y_2|$$

Detailed Proof: We verify the four axioms, relying on the properties of the absolute value function.

(i) **Non-negativity:** The absolute value of any real number is non-negative. Thus, $|x_1 - y_1| \geq 0$ and $|x_2 - y_2| \geq 0$. Their sum must also be non-negative. Thus, $d_1(x, y) \geq 0$.

(ii) **Identity of Indiscernibles:** First, assume $d_1(x, y) = 0$.

$$|x_1 - y_1| + |x_2 - y_2| = 0$$

Since both terms are non-negative, their sum can only be zero if each term is individually zero. Therefore, $|x_1 - y_1| = 0 \implies x_1 - y_1 = 0 \implies x_1 = y_1$. And, $|x_2 - y_2| = 0 \implies x_2 - y_2 = 0 \implies x_2 = y_2$. This means $x = y$.

Conversely, assume $x = y$. Then $x_1 = y_1$ and $x_2 = y_2$.

$$d_1(x, y) = |x_1 - x_1| + |x_2 - x_2| = |0| + |0| = 0$$

Thus, $d_1(x, y) = 0 \iff x = y$.

(iii) **Symmetry:** We use the property that $|a - b| = |b - a|$ for real numbers.

$$\begin{aligned} d_1(x, y) &= |x_1 - y_1| + |x_2 - y_2| \\ &= |y_1 - x_1| + |y_2 - x_2| \\ &= d_1(y, x) \end{aligned}$$

Thus, $d_1(x, y) = d_1(y, x)$.

(iv) **Triangle Inequality:** We need to prove $d_1(x, z) \leq d_1(x, y) + d_1(y, z)$. By definition, $d_1(x, z) = |x_1 - z_1| + |x_2 - z_2|$. We use the triangle inequality for real numbers: $|a + b| \leq |a| + |b|$. For the first term: $|x_1 - z_1| = |(x_1 - y_1) + (y_1 - z_1)| \leq |x_1 - y_1| + |y_1 - z_1|$. For the second term: $|x_2 - z_2| = |(x_2 - y_2) + (y_2 - z_2)| \leq |x_2 - y_2| + |y_2 - z_2|$.

Now, we sum these two inequalities:

$$\begin{aligned} d_1(x, z) &= |x_1 - z_1| + |x_2 - z_2| \\ &\leq (|x_1 - y_1| + |y_1 - z_1|) + (|x_2 - y_2| + |y_2 - z_2|) \end{aligned}$$

Rearranging the terms:

$$\begin{aligned} d_1(x, z) &\leq (|x_1 - y_1| + |x_2 - y_2|) + (|y_1 - z_1| + |y_2 - z_2|) \\ &\leq d_1(x, y) + d_1(y, z) \end{aligned}$$

Thus, the triangle inequality holds.

Since d_1 satisfies all four axioms, it is a valid metric on $\mathbb{R}^2$.

3.3 The Discrete Metric (d_0)

Definition: The Discrete metric $d_0 : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined as:

$$d_0(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

Detailed Proof: We verify the four axioms.

- (i) **Non-negativity:** The function's only possible outputs are 0 and 1. Both are non-negative. Thus, $d_0(x, y) \geq 0$.
- (ii) **Identity of Indiscernibles:** This is given directly by the definition. $d_0(x, y) = 0$ if and only if $x = y$.
- (iii) **Symmetry:** We must show $d_0(x, y) = d_0(y, x)$.
 - Case 1: $x = y$. Then $y = x$. $d_0(x, y) = 0$ and $d_0(y, x) = 0$. So $d_0(x, y) = d_0(y, x)$.
 - Case 2: $x \neq y$. Then $y \neq x$. $d_0(x, y) = 1$ and $d_0(y, x) = 1$. So $d_0(x, y) = d_0(y, x)$.

In all cases, symmetry holds.

- (iv) **Triangle Inequality:** We need to prove $d_0(x, z) \leq d_0(x, y) + d_0(y, z)$. We can analyze this by cases.
 - Case 1: $x = z$. Then $d_0(x, z) = 0$. The right side is $d_0(x, y) + d_0(y, z)$. Since both terms are non-negative, their sum is ≥ 0 . So $0 \leq d_0(x, y) + d_0(y, z)$ is always true.
 - Case 2: $x \neq z$. Then $d_0(x, z) = 1$. We must prove $1 \leq d_0(x, y) + d_0(y, z)$. Since $x \neq z$, the point y cannot be equal to both x and z . This means at least one of the following must be true: $y \neq x$ or $y \neq z$.
 - If $y \neq x$, then $d_0(x, y) = 1$.
 - If $y \neq z$, then $d_0(y, z) = 1$.The sum $d_0(x, y) + d_0(y, z)$ must be at least 1 (it could be 1+0, 0+1, or 1+1). Therefore, $1 \leq d_0(x, y) + d_0(y, z)$ is always true when $x \neq z$.

In all cases, the triangle inequality holds.

Since d_0 satisfies all four axioms, it is a valid metric on $\mathbb{R}^2$.

4 Examples of Distance Calculations

To illustrate the differences between these metrics, let's calculate the distance between two points, $P = (1, 2)$ and $Q = (4, 6)$, using each metric.

4.1 Example: Euclidean Distance

Using the d_2 metric:

$$\begin{aligned}d_2(P, Q) &= \sqrt{(4-1)^2 + (6-2)^2} \\&= \sqrt{3^2 + 4^2} \\&= \sqrt{9 + 16} \\&= \sqrt{25} = 5\end{aligned}$$

The Euclidean distance is 5.

4.2 Example: Taxicab Distance

Using the d_1 metric:

$$\begin{aligned}d_1(P, Q) &= |4-1| + |6-2| \\&= |3| + |4| \\&= 3 + 4 = 7\end{aligned}$$

The Taxicab distance is 7.

4.3 Example: Discrete Distance

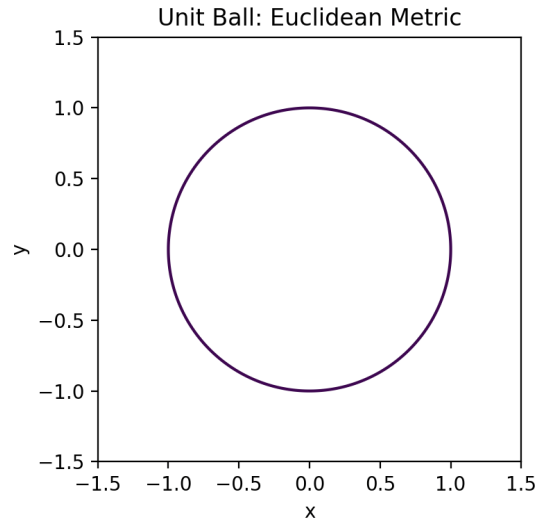
Using the d_0 metric:

$$d_0(P, Q) = \begin{cases} 0 & \text{if } (1, 2) = (4, 6) \\ 1 & \text{if } (1, 2) \neq (4, 6) \end{cases}$$

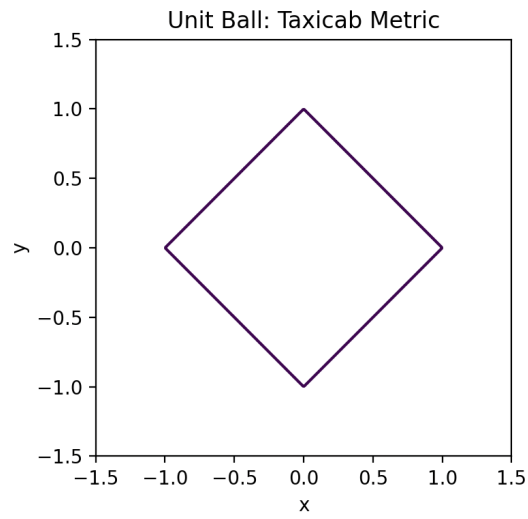
Since the points P and Q are not identical, the distance is 1.

5 Unit Balls in Different Metrics

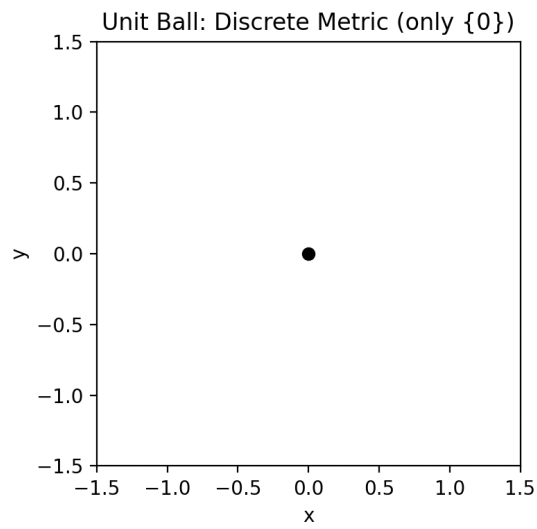
In this section, we visualize the **unit ball** (the set of all points at distance ≤ 1 from the origin) under each metric. The shape of the unit ball illustrates how the geometry of $\mathbb{R}^2$ changes with the chosen distance function.



(a) Euclidean (d_2)



(b) Taxicab (d_1)



(c) Discrete (d_0)

Figure 1: Comparison of unit balls under different metrics.

6 Python Code Used for Plotting

The following image contains the Python code used to generate the unit ball plots for the Euclidean, Taxicab, and Discrete metrics.

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-1.5, 1.5, 400)
y = np.linspace(-1.5, 1.5, 400)
X, Y = np.meshgrid(x, y)

euclid = np.sqrt(X**2 + Y**2)
taxicab = np.abs(X) + np.abs(Y)

plt.figure(figsize=(15, 4))

plt.subplot(1, 3, 1)
plt.contour(X, Y, euclid, levels=[1])
plt.title("Unit Ball: Euclidean Metric")
plt.xlabel("x")
plt.ylabel("y")
plt.gca().set_aspect("equal")

plt.subplot(1, 3, 2)
plt.contour(X, Y, taxicab, levels=[1])
plt.title("Unit Ball: Taxicab Metric")
plt.xlabel("x")
plt.ylabel("y")
plt.gca().set_aspect("equal")

plt.subplot(1, 3, 3)
plt.scatter(0, 0, color='black')
plt.title("Unit Ball: Discrete Metric (only {0})")
plt.xlabel("x")
plt.ylabel("y")
plt.gca().set_aspect("equal")
plt.xlim(-1.5, 1.5)
plt.ylim(-1.5, 1.5)
plt.show()
```

Figure 2: Python code for plotting unit balls in different metrics.

7 Analysis of How Different Metrics Affect Topological Properties

We compare how the Euclidean, taxicab, and discrete metrics influence openness, connectedness, convergence, and compactness. The behaviour follows from how their unit balls appear in Figure 1(a), Figure 1(b), and Figure 1(c).

7.1 Euclidean Metric

Openness A set is open when each point has some radius $r > 0$ such that all points within Euclidean distance less than r lie in the set. Thus openness is determined entirely by the standard Euclidean neighbourhoods.

Connectedness The plane cannot be split into two disjoint nonempty open sets, so it is connected. Because straight-line segments always remain inside the space, any two points can be joined by a continuous path. Hence the space is connected and path-connected. Any subset inherits its connectedness from the usual Euclidean topology.

Convergence A sequence converges to a point if, for every $\varepsilon > 0$, its terms eventually lie within Euclidean distance less than ε of the point. Thus convergence coincides with the usual Euclidean notion.

Compactness Compact sets are exactly the closed and bounded sets under this metric.

7.2 Taxicab Metric

Openness Although unit balls have a different shape, the same openness condition holds: each point in an open set has an ε -neighbourhood entirely contained in the set. Since taxicab and Euclidean neighbourhoods fit inside one another, the open sets induced by the two metrics are identical.

Connectedness Because the induced topologies are the same, the connectedness properties are unchanged. The space remains connected and path-connected, with continuous paths achievable through taxicab geometry just as in Euclidean geometry.

Convergence A sequence converges under the taxicab metric precisely when it converges under the Euclidean metric, since neighbourhood systems coincide.

Compactness Compact sets are the same as in the Euclidean metric.

7.3 Discrete Metric

Openness In the discrete metric, every point has a neighbourhood containing only that point. Thus every singleton is open, and therefore every subset of the space is open.

Connectedness Any set with more than one point can be written as a union of disjoint open singletons, so the only connected sets are single points.

Convergence A sequence converges only if it eventually becomes constant, since that is the only way to remain inside every neighbourhood of the limit point.

Compactness A set is compact exactly when it is finite.

7.4 Summary

- Euclidean and taxicab metrics give the same topology and therefore agree on openness, connectedness, convergence, and compactness.
- The discrete metric gives a much finer topology where every set is open, the space is totally disconnected, convergence requires eventual constancy, and only finite sets are compact.

References

- [1] W. Rudin, *Principles of Mathematical Analysis*. McGraw-Hill, 1976.
- [2] R. Bartle and D. Sherbert, *Introduction to Real Analysis*. Wiley, 2011.